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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours) /MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

THIRD SEMESTER

Discipline Specific Core Course (DSC)- MG3DSCECO200-MICROECONOMICS – I

(2024 ADMISSION ONWARDS)

SCHEME

Part A

Multiple Choice Question. Answer all questions. Each question carries 1 marks

1. b) John Hicks
2. a) Minimum cost for different outputs.
3. c) Allows for empirical measurement of utility in real world settings.
4. b) Economic region of production
5. a) the demand for the good increases
6. c) Can be purchased with a given budget
7. a) Labour markets
8. b) Labour markets
9. a) Px.Qx+Py.Qy=M
10. c) Marginal products of inputs

[1x10 = 10]

Part B

Short Answer
Answer any 10 questions
Each question carries 3 marks

11. **State the reason behind the convexity of an indifference curve.**

An indifference curve (IC) is convex to the origin because of the Diminishing Marginal Rate of Substitution () between two goods. As a consumer substitutes more units of good for good while maintaining the same level of total utility, the willingness to give up additional units of good for one more unit of good continuously diminishes.

- Mathematically, .

Due to the *Law of Diminishing Marginal Utility*, as consumption of increases, falls and increases, making the absolute slope of the IC flatter (i.e., convex).

12. **How do economies of scale and diseconomies of scale affect the long-run average cost curve in traditional analysis?**

Economies of Scale (Falling Segment): As output expands in the long run, a firm experiences real and financial economies of scale (e.g., labour specialisation, technical efficiencies, bulk purchasing discounts). This leads to a decline in per-unit cost of production, causing the Long-Run Average Cost (LAC) curve to slope downward.

Constant Returns to Scale: At an optimum scale of production, the firm experiences constant returns to scale, where increasing output does not change the average cost. At this point, the LRAC reaches its minimum.

Diseconomies of Scale (Rising Segment): Beyond the optimal scale of production (Minimum Efficient Scale), further expansion results in managerial inefficiencies, coordination failures, and communication bottlenecks. Per-unit costs begin to rise, causing the LAC curve to slope upward.

Net Effect: The interplay between economies and diseconomies of scale gives the traditional LAC curve its characteristic U-shape.

13. **Explain why for a monopolist.**

A monopolist is a price maker and faces a downward-sloping market demand curve (). To sell an additional unit of output, the monopolist must reduce the price not just for the additional unit, but for all preceding units being sold. Consequently, the Marginal Revenue () curve lies strictly below the Average Revenue () curve at all positive output levels.

14. **Define Price leadership. Who usually becomes the price leader in a market?**

Price leadership is a form of collusive oligopoly (informal/tacit collusion) where one firm sets the price for the market, and other rival firms follow by charging the same or a proportionally adjusted price.

Who becomes the Price Leader?

Dominant Firm: A dominant firm holding a major share of the total market output due to lower costs or superior resources.

Low-Cost Firm: A firm that enjoys the lowest cost of production and sets a lower price that other firms must match to survive.

Barometric Firm: An old, reputed, or well-informed firm that acts as a barometer of market condition dynamics (changes in demand/costs) without necessarily being the largest producer.

15. **Illustrate the second-degree price discrimination with real-life examples.**

Second-degree price discrimination occurs when a seller charges different per-unit prices for different quantities or blocks of the same product or service, without individual customer profiling. The firm sets a price schedule (block pricing/quantity discounts) where the price per unit decreases as the quantity consumed increases, capturing a portion of consumer surplus.

Real-Life Examples:

Utility Tariffs (Electricity/Water): Lower rate per unit for the first 100 kWh, and different tiered rates for subsequent blocks of usage.

Bulk Purchases: "Buy 2, Get 1 Free" or wholesale discount pricing (e.g., per single unit vs. per unit if bought in packs of 10).

Data Plans / Telecom Services: Tiered broadband plans based on data volume consumed.

16. **Discuss the different types of oligopoly.**

Pure / Perfect vs. Differentiated / Imperfect Oligopoly:

- *Pure:* Firms produce homogeneous/identical products (e.g., steel, cement).
- *Differentiated:* Firms produce heterogeneous/differentiated products (e.g., automobiles, soft drinks).

Open vs. Closed Oligopoly:

- *Open:* New firms are free to enter the industry.
- *Closed:* Entry of new firms is strictly blocked by legal, technical, or natural barriers.

Collusive vs. Non-Collusive Oligopoly:

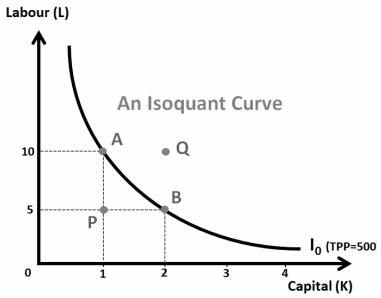
- *Collusive:* Firms cooperate to set prices/output (e.g., Cartels, Price Leadership).
- *Non-Collusive:* Firms act independently and compete aggressively (e.g., Cournot, Kinked Demand Curve model).

17. **With a diagram, show the shape of an isoquant.**

An isoquant shows all combinations of two inputs (e.g., Labour and Capital) that yield the same level of output.

Shape Characteristics:

Convex to the origin due to diminishing Marginal Rate of Technical Substitution (). Downward sloping from left to right (negative slope). Isoquants never intersect, and higher isoquants indicate higher levels of output.



Moving from point A to B along the isoquant, as more Labour () is employed, less Capital () is required to maintain the output level .

18. **What is the role of entry and exit in perfect competition and monopoly?**

Perfect Competition	Monopoly
Free entry and exit in the long run.	Blocked entry (strong legal, technological, or natural barriers).
Supernormal profits attract new entry; losses trigger exit. In long-run equilibrium, firms earn normal profit only ().	Monopolist can earn persistent supernormal profits () even in the long run.
Perfect mobility of factors of production.	Restricted factor mobility and exclusive resource ownership.

In perfect competition, firms enjoy free entry and exit. Entry of new firms eliminates supernormal profits, while exit of loss-making firms restores normal profits in the long run. In monopoly, barriers to entry prevent new firms from entering the market, allowing the monopolist to earn supernormal profits even in the long run. Thus, entry and exit promote competition in perfect competition but are restricted in monopoly.

19. **What are the features of monopolistic competition?**

- Large Number of Sellers and Buyers: Many small firms operating independently in the industry.
- Product Differentiation: Products sold by different firms are close substitutes but differentiated through brand names, packaging, quality, or service.
- Free Entry and Exit: Firms can enter or leave the industry in the long run without major legal barriers.
- Selling Costs: Heavy expenditure on advertisement, branding, and sales promotion to persuade buyers.
- Non-Price Competition: Firms compete on product features, warranty, and customer service alongside price.
- Partial Control over Price: Each firm has a downward-sloping demand curve due to brand loyalty.

20. **What are the conditions of consumer equilibrium in ordinal analysis?**

According to Hicksian Indifference Curve Analysis, consumer equilibrium requires two main conditions:

- First-Order Condition (Tangency Condition):
    - The budget line must be tangent to the highest possible indifference curve.
    -
    -
- Second-Order Condition (Convexity Condition):
    - The Indifference Curve must be strictly convex to the origin at the point of equilibrium (i.e., must be diminishing at the point of tangency).
    - At this point, the consumer attains the highest possible satisfaction with the given income and prices.

21. **Explain pragmatic approach.**

According to A. Koutsoyiannis, the *Pragmatic Approach to Demand Analysis* abandons abstract utility-maximizing assumptions of individual consumer behaviour. It accepts the empirical Law of Demand on trust and directly formulates and estimates aggregate market demand functions using observed market/statistical data. It analyses economic problems using real-world facts, experience, observation, and empirical data to arrive at effective decisions. It is widely used in business and policy-making because it provides practical solutions to real economic problems.

- Features:
    1. Focuses on group/market behaviour rather than single consumer choice.
    2. Uses multivariate functional forms estimated via econometric techniques:

    3. Formulates real-world relations directly (e.g., constant elasticity demand functions, dynamic demand functions) to solve practical business problems.
    4. Flexible and adaptable to changing economic conditions.
    5. Helps managers and policymakers make effective decisions.

22. **What is meant by Average Fixed Cost (AFC)? Why does AFC continuously fall with larger output?**

Average Fixed Cost (AFC) is the total fixed cost () per unit of output produced ().

Reason for Continuous Decline:

- Total Fixed Cost () remains constant regardless of the output level in the short run.
- As output () increases, the constant numerator () is divided by a progressively larger denominator ().
- This phenomenon is called "spreading the overhead costs."

Graphically, the AFC curve is a Rectangular Hyperbola, approaching both axes asymptotically but never touching either axis ().

23. **Evaluate whether it is rational for a perfectly competitive firm to continue production when .**

It is NOT rational for a firm to continue production if .

In the short run, a firm compares the market price (P) with its Average Variable Cost (AVC). When , Total Revenue () is insufficient to cover even the variable operating costs ().

If the firm produces: Total Loss .

If the firm shuts down: Loss only.

Since operating generates losses greater than fixed costs, a rational firm will minimize losses by shutting down immediate operations in the short run. Point where is the Shutdown Point. Continuing production increases losses, whereas shutting down limits the loss to fixed costs

alone. Therefore, according to the shutdown rule, a perfectly competitive firm should temporarily cease production when Price is less than Average Variable Cost (P < AVC).

24. **What are the conditions for profit maximisation under monopolistic competition?**

A firm under monopolistic competition maximizes its profit (or minimizes loss) when two conditions are satisfied simultaneously:

- First-Order / Necessary Condition:
  -
  - At this output, profit is maximised because the additional revenue from selling one more unit equals the additional cost of producing it.
- Second-Order / Sufficient Condition:
  -
  - (i.e., Slope of Slope of at the point of equilibrium).

  This ensures that the equilibrium is a maximum profit point, not a minimum.

After determining the profit-maximising output where MR = MC, the firm charges the price indicated by the Average Revenue (AR) or demand curve corresponding to that output. In the long run, freedom of entry ensures , so firms earn only normal profits.

25. **What happens to demand when the price of a substitute decreases?**

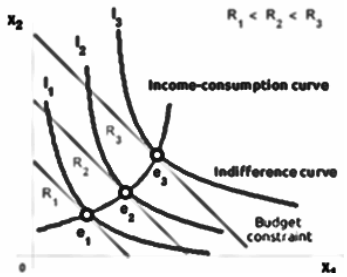
Substitutes are goods that satisfy similar consumer wants (e.g., Tea and Coffee). When the price of good (Coffee) decreases, good becomes relatively cheaper than good (Tea). Consumers shift their consumption from good to good (Substitution Effect). The demand for good decreases. Graphically, the demand curve for good shifts leftward. Thus, there is an inverse relationship between the price of a substitute and the demand for the original good. Cross elasticity of demand between substitute goods is positive ().

Part C: Short Essay

Answer any 5 questions
Each question carries 6 marks

26. **Draw an income consumption curve and give explanation.**

The Income Consumption Curve (ICC) traces the locus of optimal consumption bundles (consumer equilibrium points) as consumer income changes, holding relative prices () and consumer preferences constant.



IC , IC , and IC are successive indifference curves representing higher levels of satisfaction. E , E , and E are equilibrium points corresponding to increasing income. The curve joining these equilibrium points is the Income Consumption Curve (ICC). Start with initial budget line tangent to IC at . As income rises, budget lines shift parallelly outward (). Tangency points with higher ICs () are connected to form the ICC. The consumer reaches higher indifference curves and enjoys a higher level of satisfaction. The locus of these new equilibrium points forms the Income Consumption Curve.

When both goods are normal, the ICC slopes upward to the right (positive slope), indicating that consumption of both and increases as income increases. If is an inferior good, the ICC bends leftward / towards the Y-axis at higher income levels. If is an inferior good, the ICC bends downward / towards the X-axis.

27. **Explain the Modern Theory of Short-Run Costs. How does reserve capacity influence the AVC and SATC curves?**

Unlike traditional theory (which assumes U-shaped short-run average cost curves), the modern theory asserts that firms design plants with built-in reserve capacity to handle demand fluctuations flexibly. Because of this reserve capacity, short-run cost curves differ from the sharply U-shaped curves of the traditional theory.

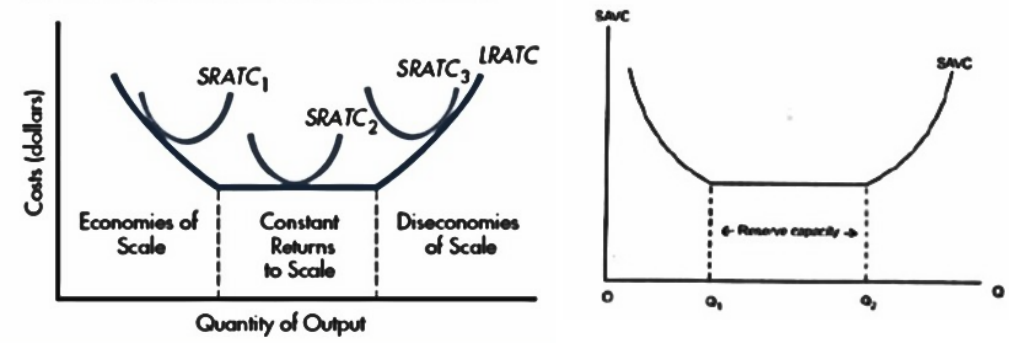
- Flat / Saucer-Shaped Average Variable Cost (SAVC):

In the traditional theory, the AVC curve is sharply U-shaped. In the modern theory, reserve capacity makes the AVC curve saucer-shaped (flat-bottomed). As output expands, SAVC falls initially due to higher efficiency. Once normal plant capacity is reached, SAVC remains flat

(constant) over a wide range of output because of reserve capacity. Only when output exceeds absolute full capacity does SAVC begin to turn sharply upward.

- Short-Run Average Total Cost (SATC):

SATC continues to decline over the flat range of SAVC because Average Fixed Cost (SAFC) continues to fall. SATC reaches its minimum at the absolute full capacity point. It has a broad flat portion, indicating that average total cost remains nearly constant over a considerable range of output. Costs rise only when production exceeds the reserve capacity and diminishing returns begin to operate.



28. **Explain short-run equilibrium of a firm under perfect competition with supernormal profits.**

In a perfectly competitive market, a firm earns supernormal (abnormal) profits in the short run when the market price is greater than the Average Total Cost (ATC) at the profit-maximising level of output.

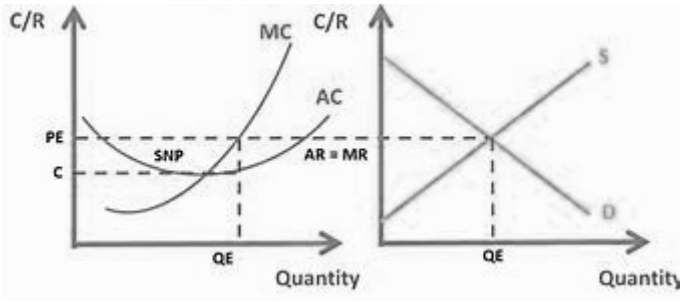
- Equilibrium Conditions:
    1. (Necessary Condition).
    2. curve intersects curve from below (Sufficient Condition).
    3. Under perfect competition, .
- **Condition for Supernormal Profit:**

  Supernormal profit occurs in the short run when Market Price () exceeds Short-Run Average Cost () at the equilibrium output (). If, at this output,

  the firm earns supernormal (abnormal) profit.

    - Market price is determined by demand-supply equilibrium at .

- The firm produces where .
- At this output , project a line up to the curve to find per-unit cost ().
- The rectangle area bounded by represents total supernormal profit.



A perfectly competitive firm earns supernormal profits in the short run when the market price is greater than the Average Total Cost (ATC) at the equilibrium output determined by MR = MC. However, these profits are temporary, as free entry of new firms in the long run eliminates supernormal profits by increasing market supply.

29. **Differentiate Between Price Leadership and Cartel.**

Price Leadership is an informal form of collusion, where one firm leads and others follow its pricing decisions. A Cartel is a formal agreement among firms to fix prices and output with the objective of maximising joint profits and reducing competition.

Feature	**Price Leadership**	**Cartel**
Nature of Agreement	Tacit / Informal collusion without any written or explicit formal agreement.	Explicit / Formal collusion where firms enter a binding, written agreement.
Organizational Body	No central joint body exists; one dominant or barometric firm sets the price.	A centralized organ/board (e.g., OPEC) is established to set prices and quotas.
Independence	Member firms retain individual control over their operations and sales strategies.	Member firms surrender partial sovereignty regarding pricing and output quotas.
Legal Status	Less easily detected by anti-trust laws due to its informal/implicit nature.	Mostly illegal in many jurisdictions under anti-monopoly regulations.
Market Share	Output allocations are implicitly	Output quotas and market

Allocation	adjusted around the leader's price.	territories are explicitly divided among members.

30. **"Producer's equilibrium ensures maximum efficiency of resources." Do you agree? Give reasons.**

Yes, agreed. Producer's equilibrium represents the optimal allocation of inputs where a firm maximizes output for a given cost outlay or minimizes cost for a target output level. Marginal Rate of Technical Substitution (MRTS) equals the ratio of input prices:

where:

- = Price of labour
- = Price of capital

The isoquant must be convex to the origin, ensuring a least-cost combination of inputs.

Reasons:

1. Least-Cost Combination of Inputs: At equilibrium, the Isoquant is tangent to the Isocost line, satisfying. This condition ensures that every unit of expenditure on Labour and Capital yields equal marginal productivity, eliminating technical and allocative inefficiency. The producer chooses the combination of labour and capital that minimises production cost.
2. Elimination of Resource Waste: Any deviation from the equilibrium point implies higher costs per unit of output or lower output for the same outlay, signifying resource misallocation.
3. Optimal Scale Optimisation: It forces the producer to adjust factor proportions according to relative input prices (and), achieving full technical efficiency.
4. Profit Maximisation: Cost minimisation combined with efficient production leads to maximum profit.

31. **Explain the linear expenditure system in consumer theory.**

Developed by Richard Stone (1954), the Linear Expenditure System (LES) is an empirical model of consumer demand derived from a specific utility function (Stone-Geary utility function). It explains how a consumer allocates income between essential (subsistence) consumption and discretionary expenditure. According to the model, a consumer first spends enough to satisfy

minimum basic needs and then allocates the remaining income among goods according to fixed proportions.

Assumptions

- Consumers have a minimum (subsistence) level of consumption for each good.
- Income is divided into:
    - Subsistence expenditure, and
    - Discretionary (surplus) expenditure.
- Prices are given and consumer preferences remain unchanged.
- Mathematical Formulation:

The Linear Expenditure System is represented as:

where:

- = Quantity demanded of good
- = Minimum (subsistence) consumption of good
- = Consumer's income
- = Price of good
- = Proportion of discretionary income spent on good

Subsistence / Minimum Expenditure (): Represents the mandatory basic minimum quantity () of good required by the consumer regardless of price or income.

1. Supernumerary Income (): The discretionary income remaining after purchasing minimum subsistence amounts of all goods.
2. Marginal Propensity to Spend (): Represents the proportion of supernumerary income spent on good , where .

The Linear Expenditure System (LES) explains consumer behaviour by assuming that income is first used to satisfy minimum consumption requirements, and the remaining income is distributed among goods according to fixed expenditure shares. It provides a practical and realistic framework for analysing consumer demand and expenditure patterns. It guarantees theoretical properties like additivity, homogeneity, and symmetry, making it widely used in applied economic research.

32. Discuss how changes in input prices affect the position of isocost line and producer's equilibrium.

Effects of Changes in Input Prices:

Change in Price of Labour ():

- Decrease in (Capital price constant): The Isocost line rotates outward along the X-axis (Labour axis). The firm can buy more labour with the same budget.
- Increase in : The Isocost line rotates inward along the X-axis.

Change in Price of Capital ():

- Decrease in : The Isocost line rotates outward along the Y-axis (Capital axis).

Impact on Producer's Equilibrium:

- Substitution Effect: A change in relative input prices () changes the slope of the Isocost line.
- Shift to New Tangency Point: The firm establishes a new equilibrium where the new Isocost line is tangent to an Isoquant.
- Resource Reallocation: The firm substitutes the relatively cheaper input for the relatively more expensive input to maintain cost-efficiency.

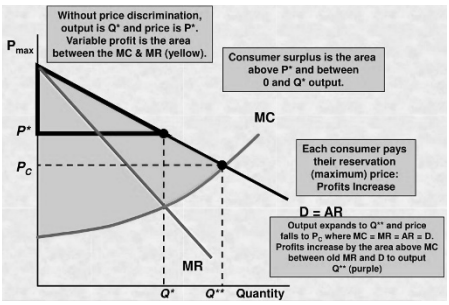
33. Show how consumer surplus changes under first-degree price discrimination.

First-Degree Price Discrimination (Perfect Price Discrimination): Occurs when a monopolist charges each consumer the maximum reservation price they are willing to pay for each unit sold.

Impact on Consumer Surplus: Under a single uniform price (), consumers pay for all units, leaving them with Consumer Surplus (CS) equal to the area below the demand curve and above .

Under First-Degree Price Discrimination: a) The monopolist extracts 100% of the Consumer Surplus. b) Consumer Surplus becomes Zero (). c) The entire consumer surplus is converted into Monopolist's Profit.

Efficiency Outcome: Output expands to where (identical to competitive output). Although total economic welfare is maximized (allocative efficiency), social distribution is highly unequal because the producer captures all gains.



34. Discuss why the firm in perfect competition is a price taker.

- Large Number of Buyers and Sellers: Individual firms produce a negligible fraction of total industry output. A single firm's output decisions cannot shift total market supply or influence price.
- Homogeneous Product: Products sold by all firms are perfect substitutes. If a firm raises its price even slightly above market price, its sales drop to zero.
- Perfect Market Information: Consumers and sellers possess complete market knowledge regarding price and quality, preventing any price dispersion.
- Horizontal Demand Curve (): The firm faces a perfectly elastic demand curve at the market-clearing price (), set by total industry demand and supply. Thus, .

35. Describe Revealed Preference theory.

Introduced by Paul Samuelson (1938) to explain consumer demand behavior empirically without using subjective notions like utility or indifference curves. Consumer preferences are "revealed" through observed choices in the market under given budget constraints.

Key Axioms:

- Strong Ordering: The consumer selects a unique bundle of goods out of all affordable bundles available within their budget constraint.
- Weak Axiom of Revealed Preference (WARP): If bundle is chosen over bundle when both are affordable (is revealed preferred to), then can never be chosen over at any other price-income situation where is also affordable.

Samuelson derived the Law of Demand directly from WARP using basic consistency assumptions, proving that a negative income effect is not required to establish a downward-sloping demand curve for normal goods. Revealed preference theory states that the consumer's product or service choice under different income and price levels provides information about

their preferences. For example, if there were two combinations of goods (A and B) and if the consumer chooses A (which is expensive) over B, then the consumer reveals their preference for A over B. The theory is based on the premise that consumers are rational thinkers who compare many options before making a decision. As a result, when a consumer selects one choice over another, it must be the one they prefer. When a consumer's consumption pattern is determined by observing the bundle of goods they choose over other alternatives, when all of them are within the budget is known as revealed preference. A budget line is a diagram that depicts the combination of products that a consumer can purchase at a certain income and price level. If bundle a is revealed preferred over bundle b in budget set B, then the WARP says that bundle b cannot be revealed preferred over bundle a in any budget set B'. This would be equally true had b been located anywhere else on or below the blue line. The bundle c will not violate WARP even if it is chosen in budget set B', because it is not on or below the blue line of affordable choices at the time of the observed choice of a.